

Linear mixed models in R

Day 2

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Syntax of a crossed mixed-effects model

RT ~ Context + (1 | Subject)



Dependent variable



Independent
variables/fixed effects



Random effects

Syntax of a crossed mixed-effects model

RT ~ Group * Context + Age + (1 + Context | Subject) + (1 | Item)



Dependent variable



Independent
variables/fixed effects



Random effects
(crossed between
Subject and Item)

Syntax of a crossed mixed-effects model

$RT \sim \text{Group} * \text{Context} + \text{Age} + (1 + \text{Context} \mid \text{Subject}) + (1 \mid \text{Item})$



Independent variables/fixed
effects

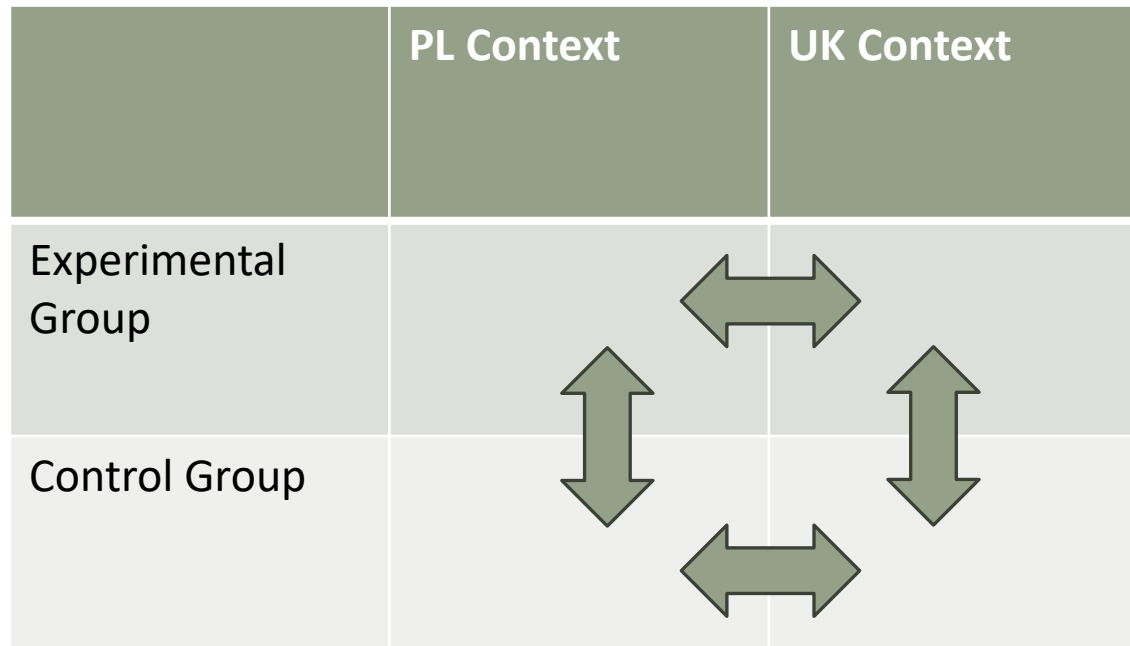
: - interaction

* - short-form for single fixed
effects and interaction between
effects; equals to:

Group + Context + Group:Context

Interactions

Group*Context = Group + Context + Group:Context



Interactions

No interaction – effects are additive

	PL Context	UK Context
Experimental Group	500ms	550ms
Control Group	480ms	530ms

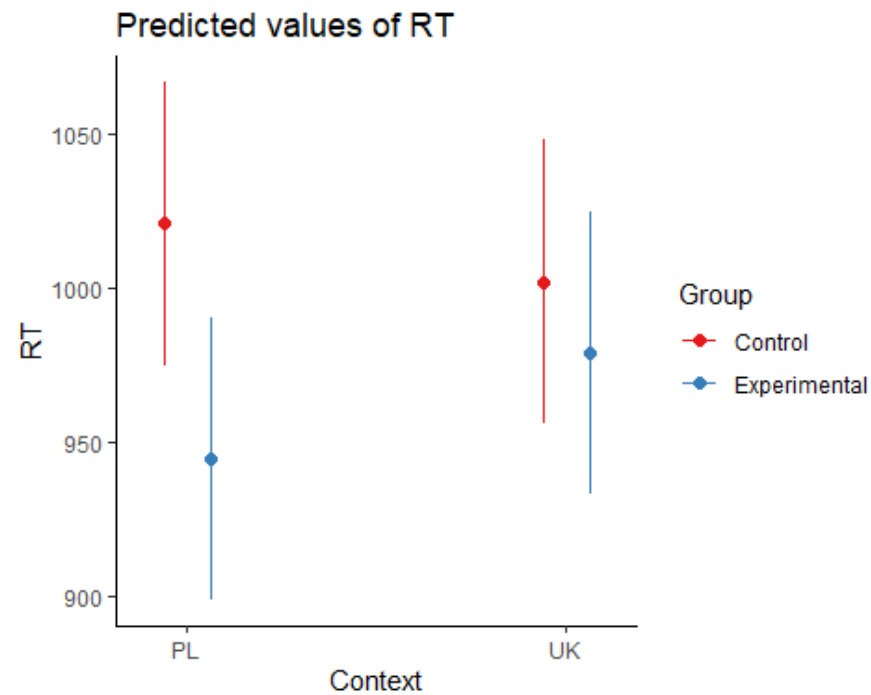
Sign. Interaction – effects are not additive

	PL Context	UK Context
Experimental Group	500ms	550ms
Control Group	480ms	480ms

Interactions

Significant interactions do not give sensible information on their own

- Additional tests are necessary to understand their effects



Syntax of a crossed mixed-effects model

RT ~ Group * Context + Age + (1 + Context | Subject) + (1 | Item)



Dependent variable



Independent
variables/fixed effects



Random effects
(crossed between
Subject and Item)

: - interaction

* - short-form for single
fixed effects and
interaction between
effects; equals:

Group + Context +
Group:Context

Crossed vs. nested random effects

Crossed: each level of one random effect occurs at each level of the other random effect, and vice versa

- e.g., each subject sees every item, and each item is seen by every subject

Nested: There are some levels of one random effect (the nested one) that only occur within one level of the other random effect (the nesting one)

- E.g. tested several students in several schools. A given student only occurs in *one* school, not in every school

A model can include both (e.g. *Students* nested under *Schools*, but *Items* fully crossed with those)

Syntax of a crossed mixed-effects model

$RT \sim \text{Group} * \text{Context} + \text{Age} + (1 + \text{Context} \mid \text{Subject}) + (1 \mid \text{Item})$

Nested random effect:

$RT \sim \text{Group} * \text{Context} + \text{Age} + (1 \mid \text{School/Student})$

Identical to:

$RT \sim \text{Group} * \text{Context} + \text{Age} + (1 \mid \text{School}) + (1 \mid \text{School:Student})$

Summary-Output of an LME

Amount of variance for each random term

Coefficients for fixed effects

Correlation of fixed effects
(NOT correlation of predictors or collinearity)

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: RT ~ Group + Context + Group:Context + Age + AoA + Trial + (1 +
## Context | Subject) + (1 + Context | ItemNr)
## Data: PN_Data
##
## Random effects:
## Groups Name Variance Std.Dev. Corr
## ItemNr (Intercept) 24021 154.99
## ContextUK 3449 58.73 -0.15
## Subject (Intercept) 19522 139.72
## ContextUK 16426 128.16 -0.41
## Residual 55688 235.98
## Number of obs: 7227, groups: ItemNr, 210; Subject, 74
##
## Fixed effects:
## Estimate Std. Error t value
## (Intercept) 1019.6555 79.0877 12.893
## GroupExperimental -59.6657 37.2576 -1.601
## ContextUK -11.2579 23.2922 -0.483
## Age -1.9501 2.4100 -0.809
## AoA 3.2835 3.9845 0.824
## Trial 0.4629 0.2875 1.610
## GroupExperimental:ContextUK 45.7348 32.1411 1.423
##
## Correlation of Fixed Effects:
## (Intr) GrpExp CntxUK Age AoA Trial
## GrpExprmntl 0.196
## ContextUK -0.127 0.290
## Age -0.810 -0.309 -0.012
## AoA -0.269 -0.217 0.001 -0.234
## Trial -0.110 -0.001 -0.006 0.000 0.001
## GrpExpr:CUK 0.088 -0.404 -0.704 0.010 -0.001 0.000
```

Random effects-Output

For each random effect, how much each subject's effect differs from the overall fixed effect

```
ranef(model_GeneralModel)
## $Subject
##      (Intercept)    ContextUK
## AF1310    34.183826    71.901098
## AG0712   -26.795064   -10.985648
## AG0911  -243.449395   170.020494
## AJ1312   -58.494452    59.235755
## AJ1611  -239.331334   327.456270
## AJ3007  -148.513248   -77.695294
## AK0807   231.764526  -167.158225
## AK1612  -210.589850    13.285782
...
```

Syntax of a crossed mixed-effects model

RT ~ Group * Context + Age + (1 + Context | Subject) + (1 | Item)



Dependent variable



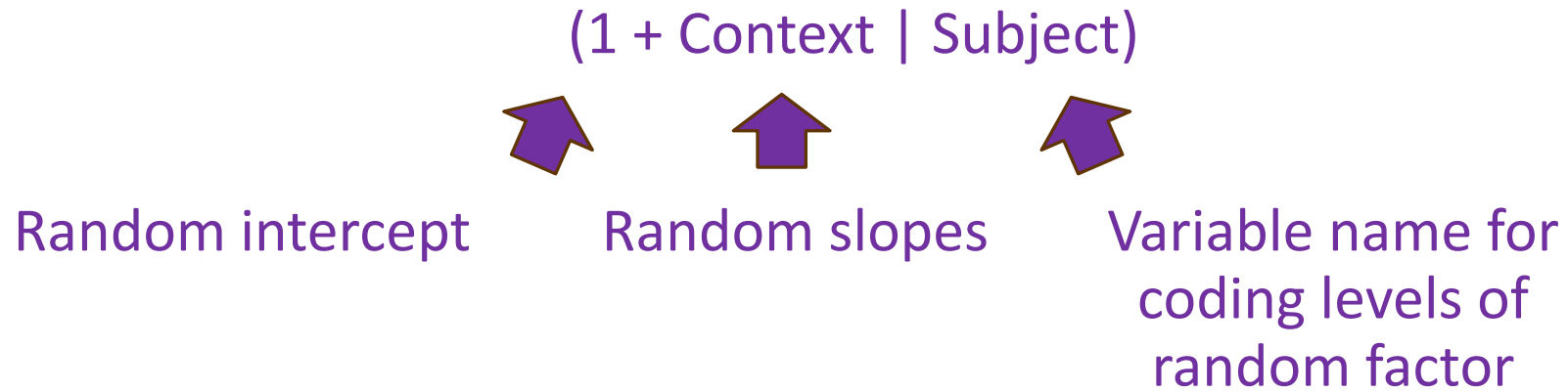
Independent
variables/fixed effects



Random effects
(crossed between
Subject and Item)

- Random effects are grouping factors
- Always categorical
- Recommended to have at least five levels

Syntax of random effects



We added subject intercepts

$$y_{iCond} = \beta_0 + \beta_{Cond}x_{iCond} + \epsilon_{Cond} + u_{0i}$$

But we can also add subject slopes

$$y_{iCond} = \beta_0 + (\beta_{Cond} + u_{1i})x_{iCond} + \epsilon_{Cond} + u_{0i}$$

β_0 -average naming latency

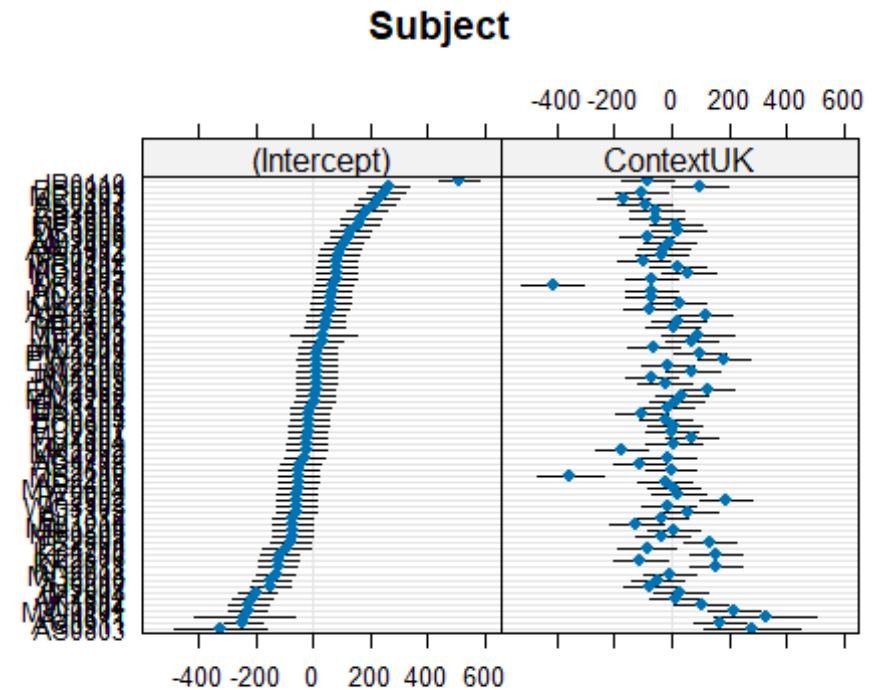
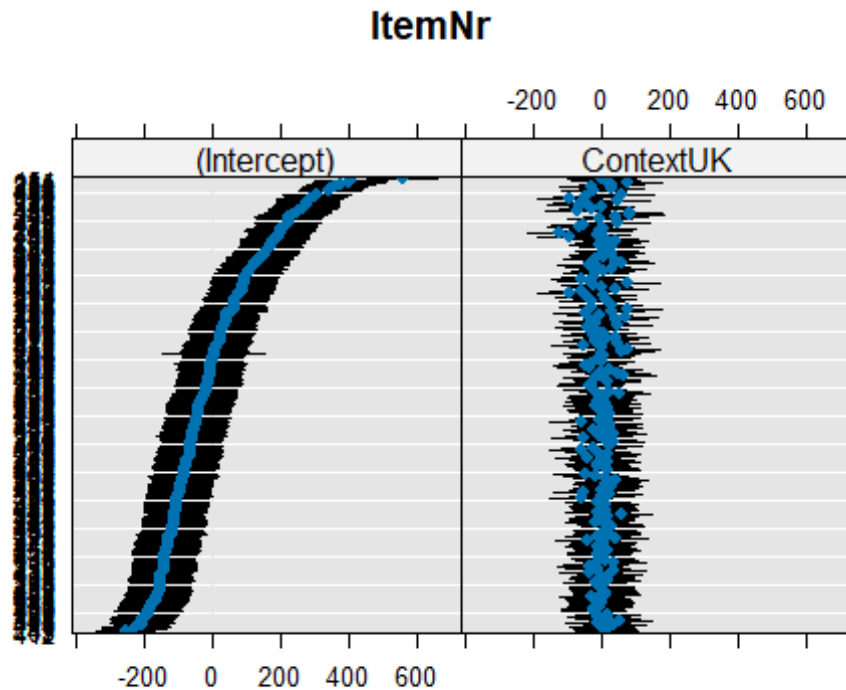
u_{0i} -subject dependant adjustment

ϵ_{Cond} -error/noise term

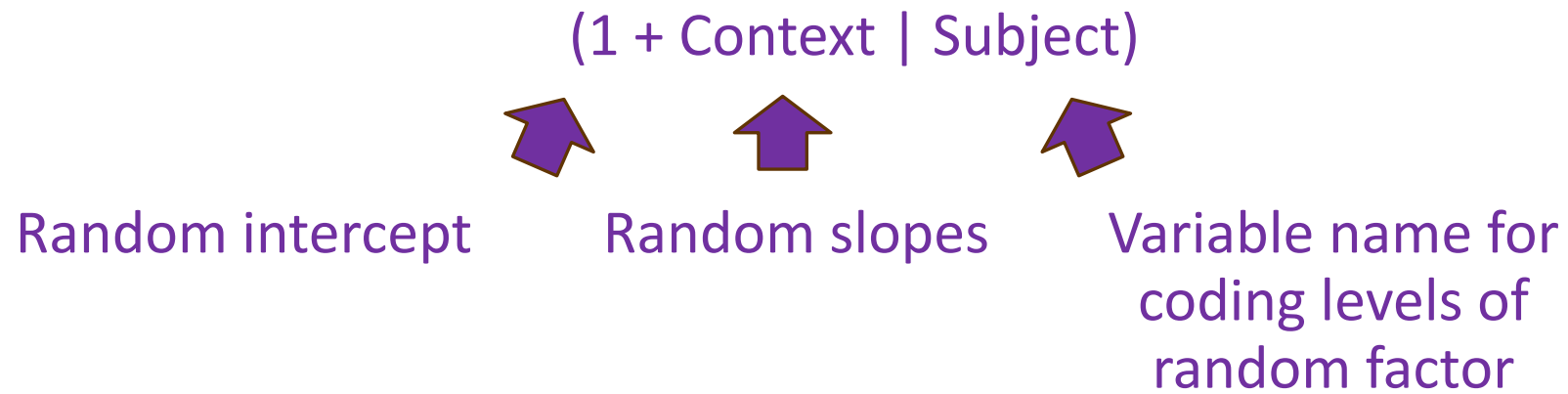
Subject adjustments for intercept and fixed effects

$RT \sim \text{Group} + \text{Context} + \text{Group}:\text{Context} + \text{Age} + \text{AoA} + \text{Trial} + (1 + \text{Context} | \text{Subject}) + (1 + \text{Context} | \text{ItemNr})$

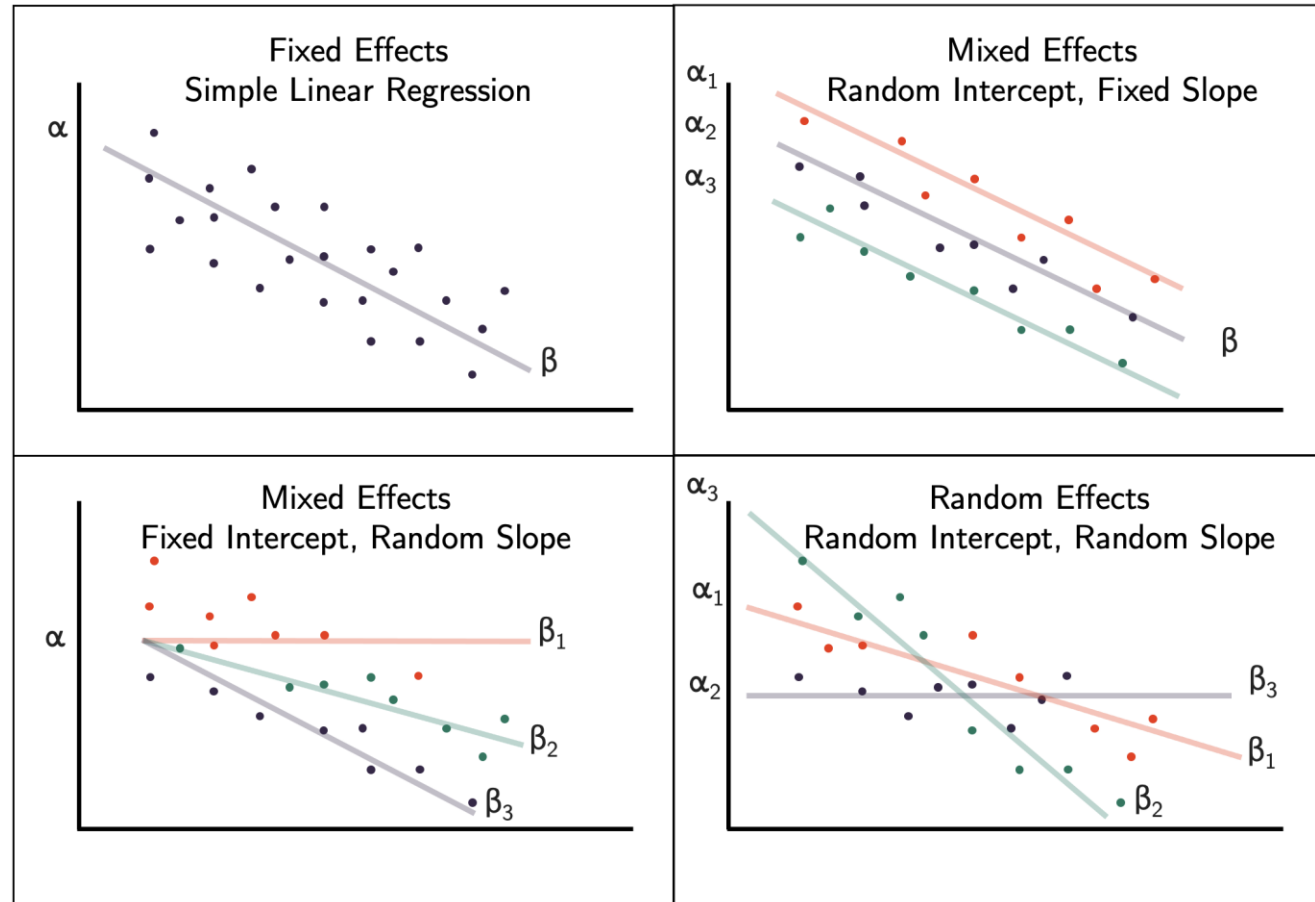
```
library(lattice)
print(dotplot(ranef(model_GeneralModel, condVar = TRUE)))
```



Syntax of random effects



Random intercept and random slope



Random intercept and no slope

```
summary(lmer(RT ~ Context + (1 | Subject) + (1 | ItemNr),  
data=PN_Data))
```

```
## Random effects:
```

##	Groups	Name	Variance	Std.Dev.
##	ItemNr	(Intercept)	23600	153.6
##	Subject	(Intercept)	15184	123.2
##	Residual		59859	244.7

Random intercept and random slope

```
summary(lmer(RT ~ Context + (1 + Context | Subject) + (1 |  
ItemNr), data=PN_Data))
```

```
## Random effects:
```

##	Groups	Name	Variance	Std.Dev.	Corr
##	ItemNr	(Intercept)	23827	154.4	
##	Subject	(Intercept)	20662	143.7	
##		ContextUK	17133	130.9	-0.46
##	Residual		56532	237.8	

No intercept and random slope

```
summary(lmer(RT ~ Context + (0 + Context | Subject) + (1 |  
ItemNr), data=PN_Data))
```

```
## Random effects:
```

##	Groups	Name	Variance	Std.Dev.	Corr
##	ItemNr	(Intercept)	23827	154.4	
##	Subject	ContextPL	20663	143.7	
##		ContextUK	20330	142.6	0.58
##	Residual		56532	237.8	

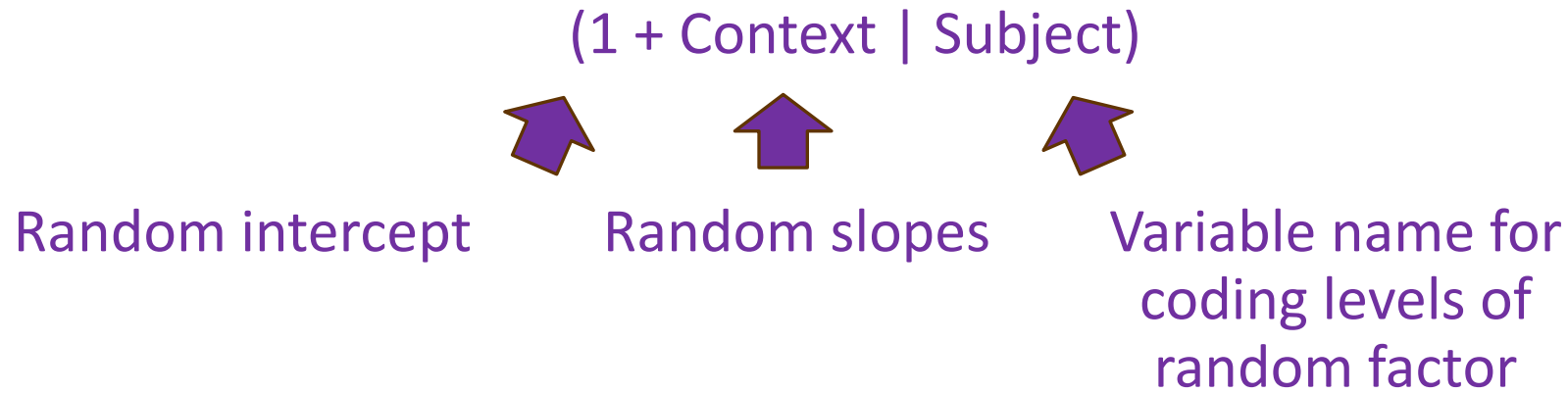
Random intercept and random slope

```
summary(lmer(RT ~ Context + (1 + Context || Subject) + (1 + Context  
| ItemNr), data=PN_Data))
```

Random effects:

##	Groups	Name	Variance	Std.Dev.	Corr
##	ItemNr	(Intercept)	24424	156.28	
##		ContextUK	3427	58.54	-0.17
##	Subject	ContextPL	8073	89.85	
##		ContextUK	7906	88.91	-0.07
##	Subject.1	(Intercept)	12515	111.87	
##	Residual		55673	235.95	

Random structure: correlations between intercept and slope



| - random intercept and random slope are correlated with each other

|| - no correlations between random effects

$(\text{Context} \parallel \text{Subject})$

This expands to:

$(0 + \text{Context} \mid \text{Subject}) + (1 \mid \text{Subject})$

Random structure: correlations between intercept and slope

$y_{iCond} = \beta_0 + (\beta_{Cond} + u_{1i})x_{iCond} + \epsilon_{Cond} + u_{0i}$

```
model_Large8 = lmer(RT ~ Group+Context + (1
+Context|Subject) +(1| ItemNr), data=PN_Data)
```

Between-subject variability in the intercept:

$$u_0 \sim Normal(0, \sigma_{u0})$$

Between-subject variability in the slope:

$$u_1 \sim Normal(0, \sigma_{u1})$$

Within-subject variability:

$$\epsilon \sim Normal(0, \sigma)$$

```
## Random effects:
```

##	Groups	Name	Variance	Std.Dev.	Corr
##	ItemNr	(Intercept)	23832	154.4	
##	Subject	(Intercept)	19854	140.9	
##		ContextUK	16898	130.0	-0.44
##	Residual		56540	237.8	

Random structure: correlations between intercept and slope

```
(1 +Context|Subject)
```

```
## Random effects:
```

##	Groups	Name	Variance	Std.Dev.	Corr
##	Subject	(Intercept)	19854	140.9	
##		ContextUK	16898	130.0	-0.44
##	Residual		56540	237.8	

```
(1 +Context||Subject)
```

```
## Random effects:
```

##	Groups	Name	Variance	Std.Dev.	Corr
##	Subject	ContextPL	3843	61.99	
##		ContextUK	4623	68.00	-1.00
##	Subject.1	(Intercept)	16011	126.53	
##	Residual		56540	237.78	

Random structure: correlations between intercept and slope

Correlations between intercepts and slopes are frequently assumed in research

- Group with higher intercept will also have a higher slope
- If they are present, they can be used to calculate better estimates
- Even low correlations can contribute to better estimates

But they make the model more complex

Correlations of 1 between slope and intercept can prevent the model from converging

- 1 or very close to one, usually means that the model failed to calculate the correlation
- Usually because there was not enough data
- Removing correlations by using `||` allows model to converge
- But it is recommended to also switch to a simpler model

Assumptions of linear mixed models

The explanatory variables are related linearly to the response and additive.

The residuals have constant variance.

The residuals are normally distributed.

The explanatory variables are related linearly to the response.

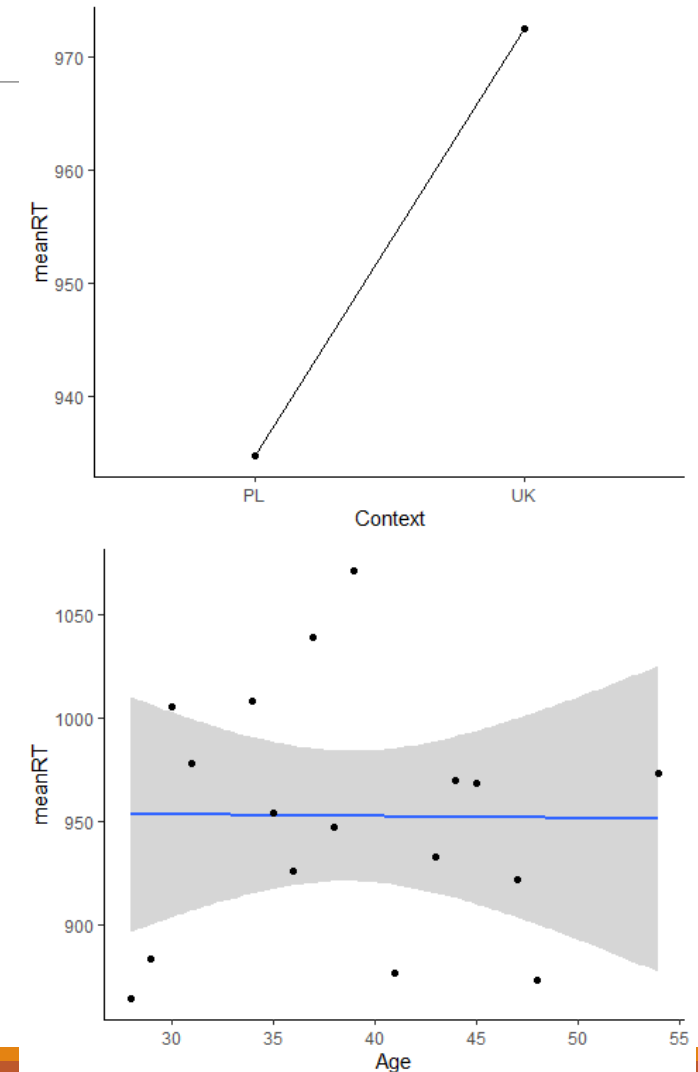
Linear mixed effects models rely on linear relationships between the variables

Categorical variables are treated as 2-level variables

Continuous variables are used in linear regression

But many effects are non-linear:

- Introduction of additional predictors and constructs to approximate non-linear effects
- Switch to Generalised-Mixed-Models and other tools



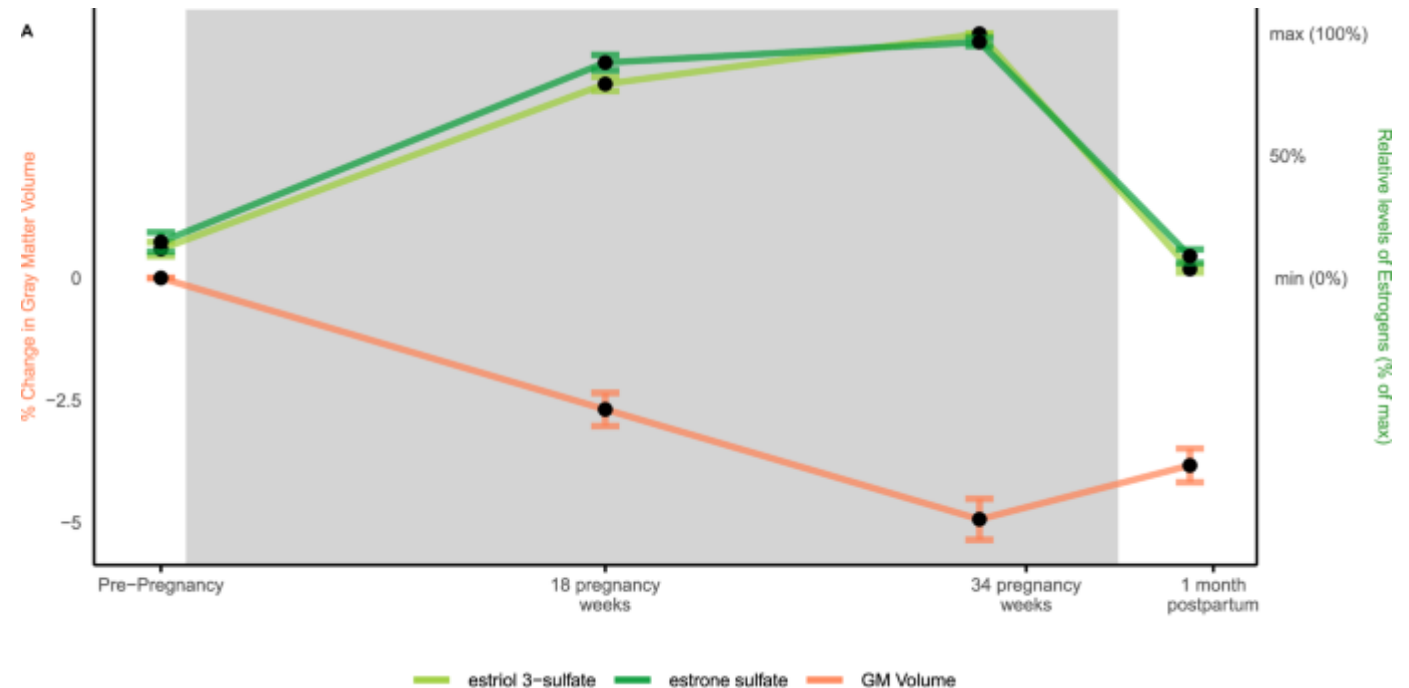
Analysing nonlinear effects with LMEs

Fixed effects in LMEs are always treated as linear

Many processes are non-linear

LMEs allow the abstraction of non-linear effects as linear ones

-> but you lose information and you could introduce erroneous effects



Servin-Barthet et al. (2025)

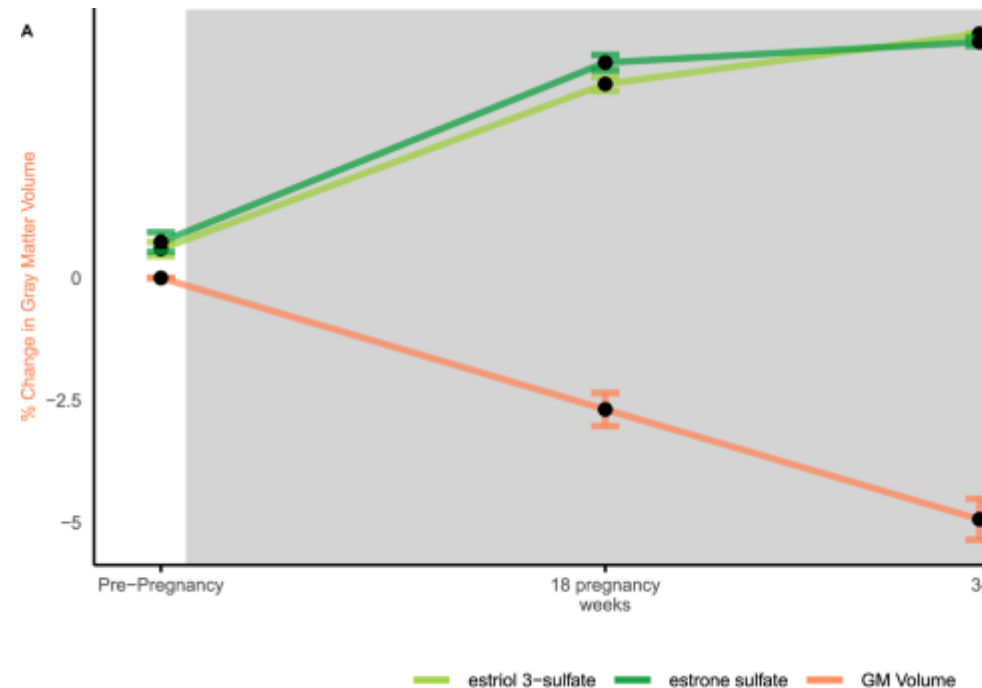
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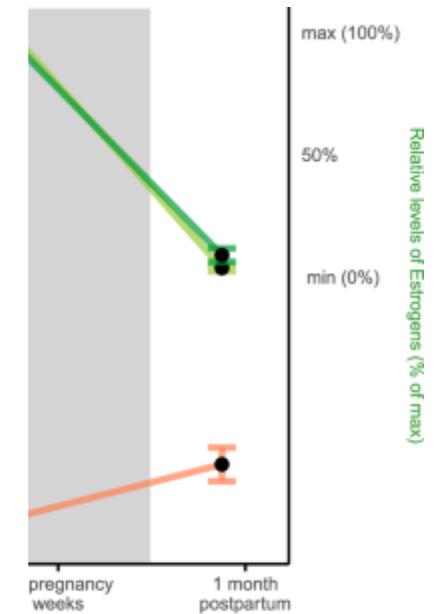
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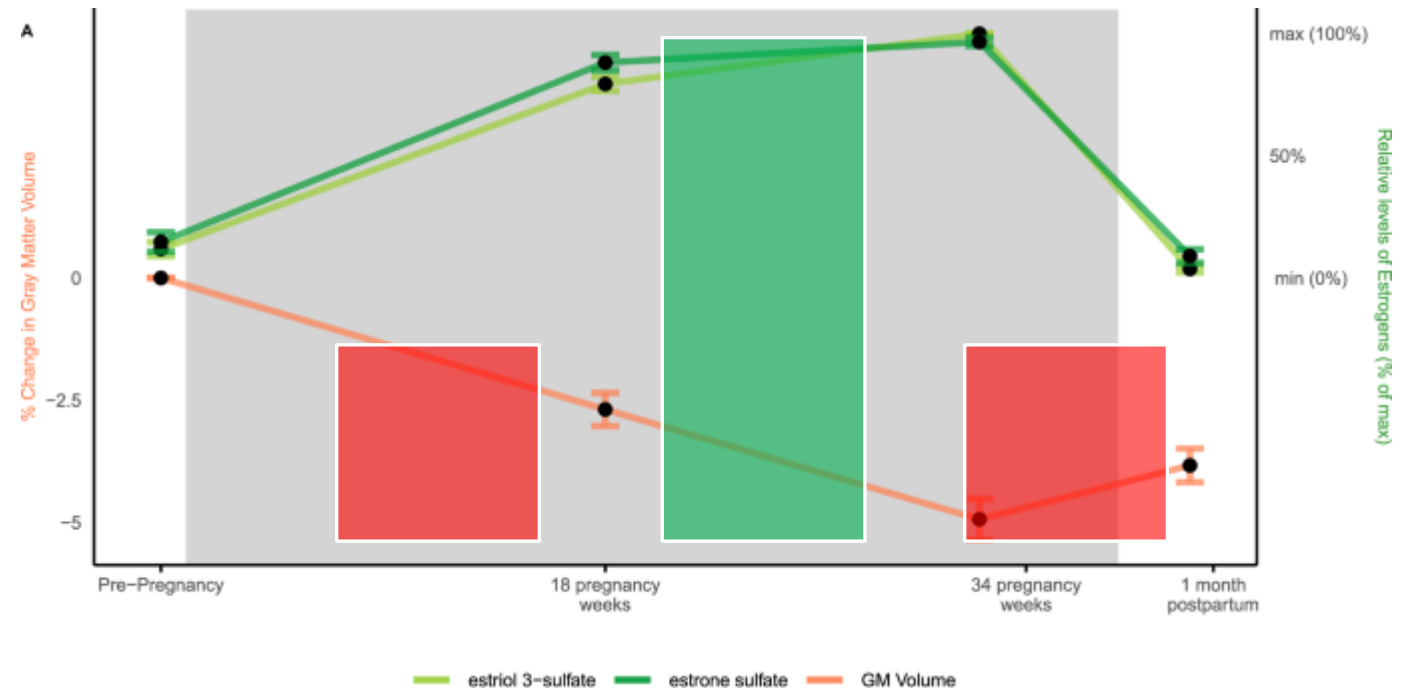
Analysing nonlinear effects with LMEs

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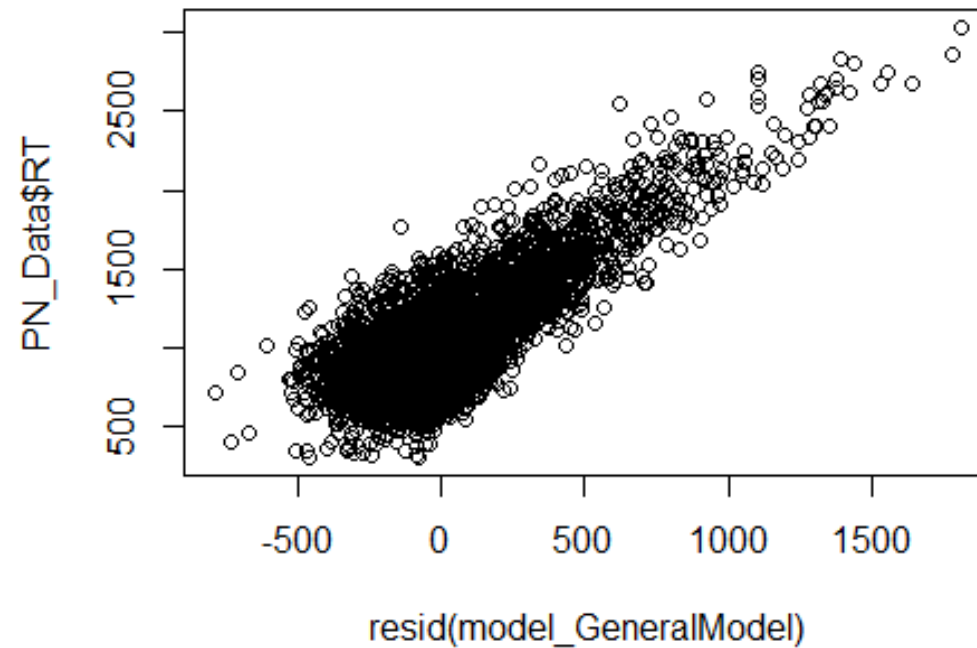
-> but you lose information and you could introduce erroneous effects



Servin-Barthet et al. (2025)

The explanatory variables are related linearly to the response.

```
plot(resid(model_GeneralModel), PN_Data$RT)
```

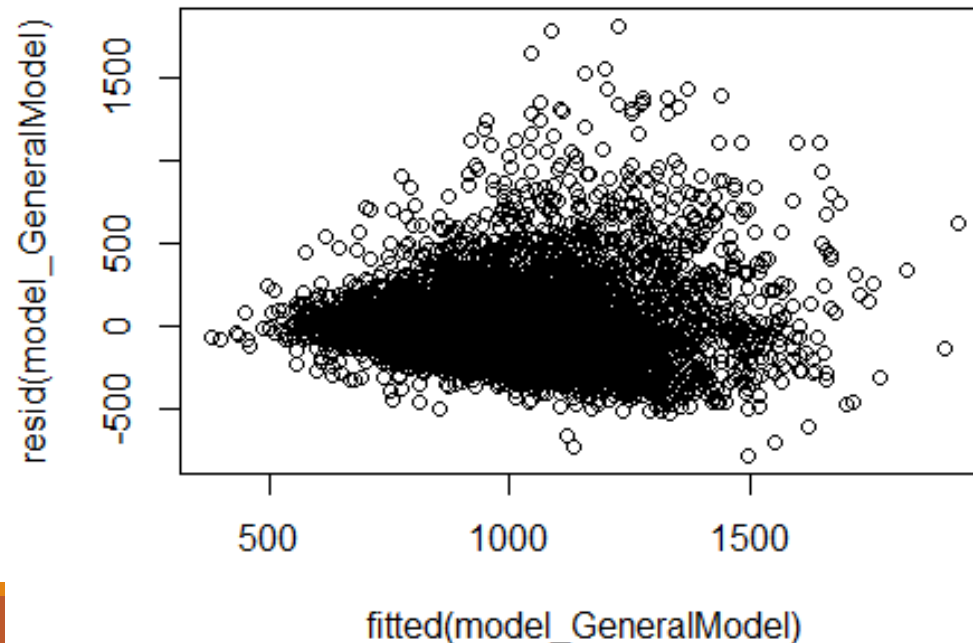


The residuals have constant variance.

$$y_{iCond} = \beta_0 + \beta_{Cond}x_{iCond} + \epsilon_{Cond} + u_{0i}$$

Plotting residuals against fitted values will indicate if there is non-constant error variance.

```
plot(fitted(model_GeneralModel), resid(model_GeneralModel))
```



Homoscedasticity

We are looking for a consistent vertical spread

-> variability should stay the same across different levels of the dependent variable

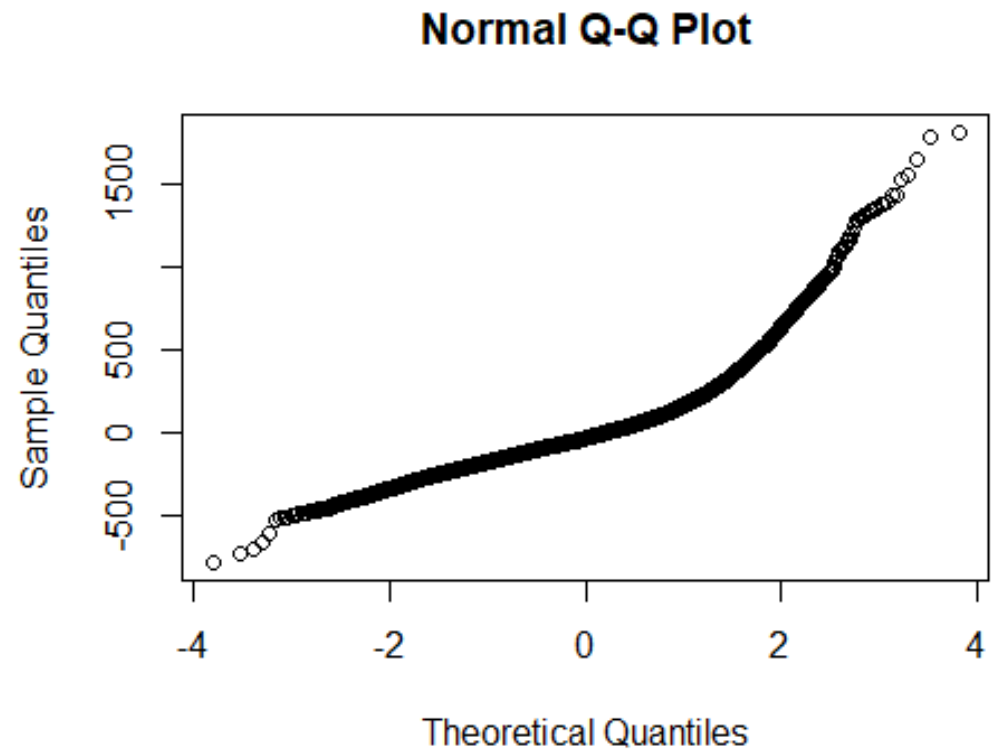
The errors are Normally distributed.

The errors are Normally distributed.

```
res_model <-  
residuals(model_GeneralModel)  
qqnorm(res_model)
```

If the line curves away from the diagonal, the normality assumption is not met.

The solution might be transformation of the data.



Transforming data

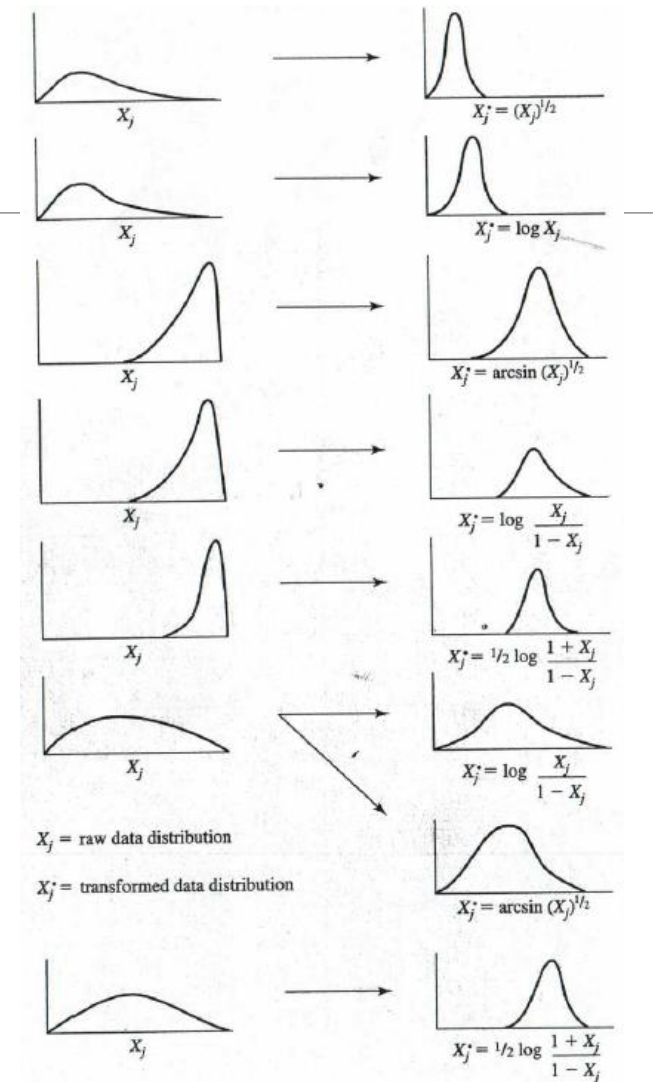
Many experimental results are not normally distributed (RT, questionnaire scales etc.)

Box-Cox procedure can be used to determine best transformation

However, transformation can also lead to bad results and a switch to GLM might be better (Lo and Andrews, 2015)

Transforming data

Data transformation is done by applying some function to the data, which changes the distribution shape, but maintains the overall data distribution



Box-Cox package

Box-Cox Transformation

- Find the exponent λ to approximate the normal distribution as close as possible

Not easily implemented for LMEs

But in linear regression

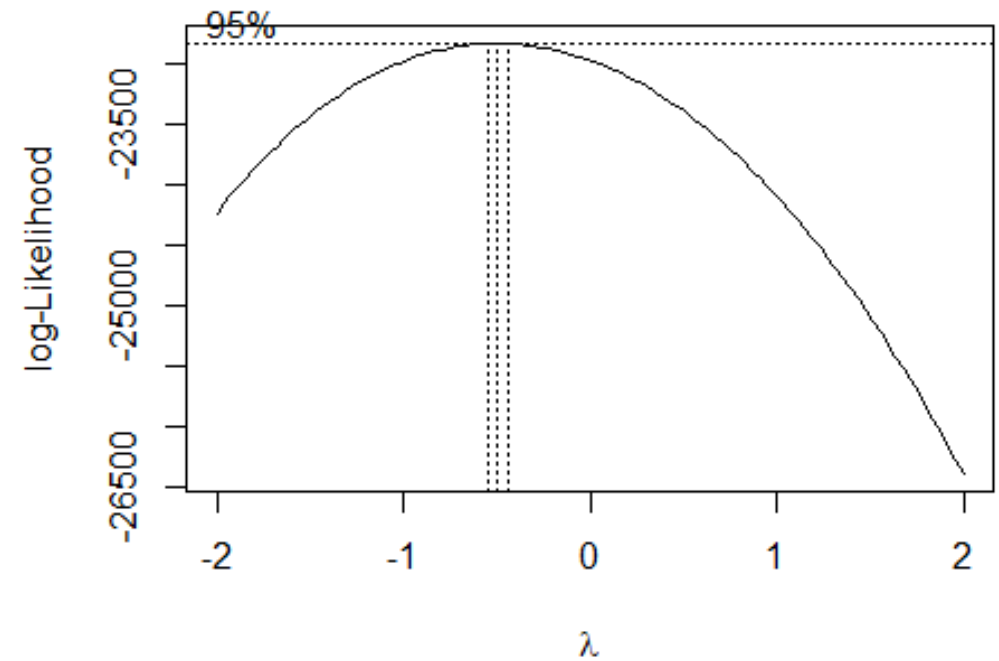
Lambda value (λ)	Transformed data (Y')
-3	$Y^{-3} = 1/Y^3$
-2	$Y^{-2} = 1/Y^2$
-1	$Y^{-1} = 1/Y$
-0.5	$Y^{-0.5} = 1/\sqrt{Y}$
0	$\log(Y)$
0.5	$Y^{0.5} = \sqrt{Y}$
1	$Y^1 = Y$
2	Y^2
3	Y^3

How to use Box-Cox package

```
library(MASS)
```

```
m0 <- lm(RT~1+Context*Group, data  
= PN_Data)
```

```
boxcox(m0)
```

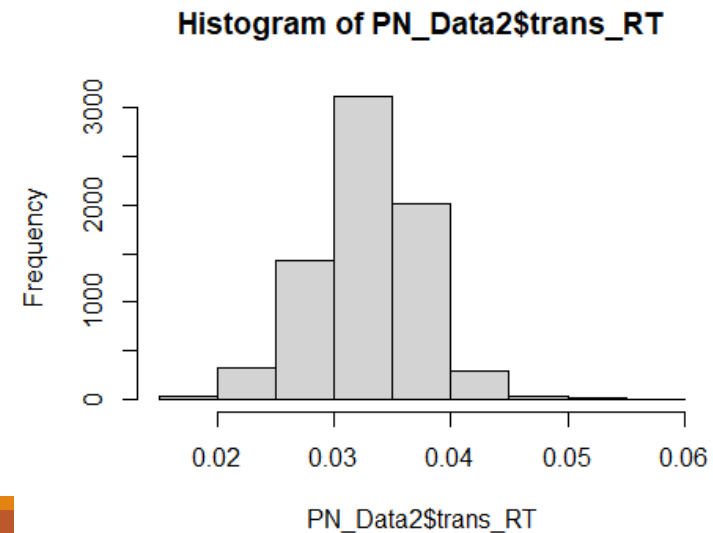
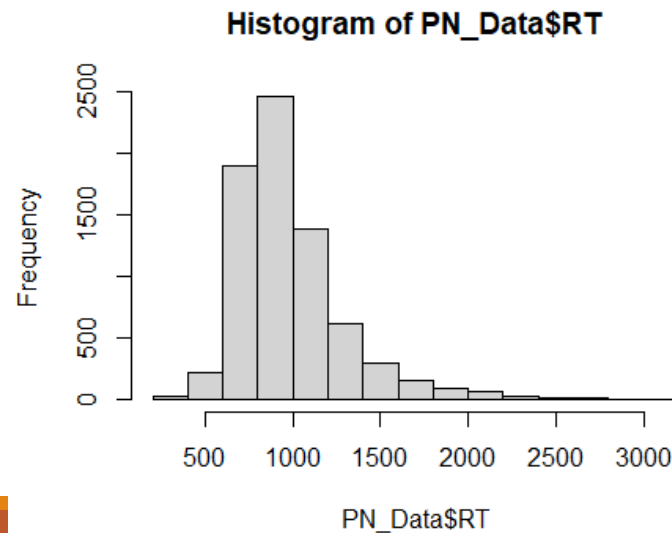


Transforming reaction times

```
hist(PN_Data$RT)
```

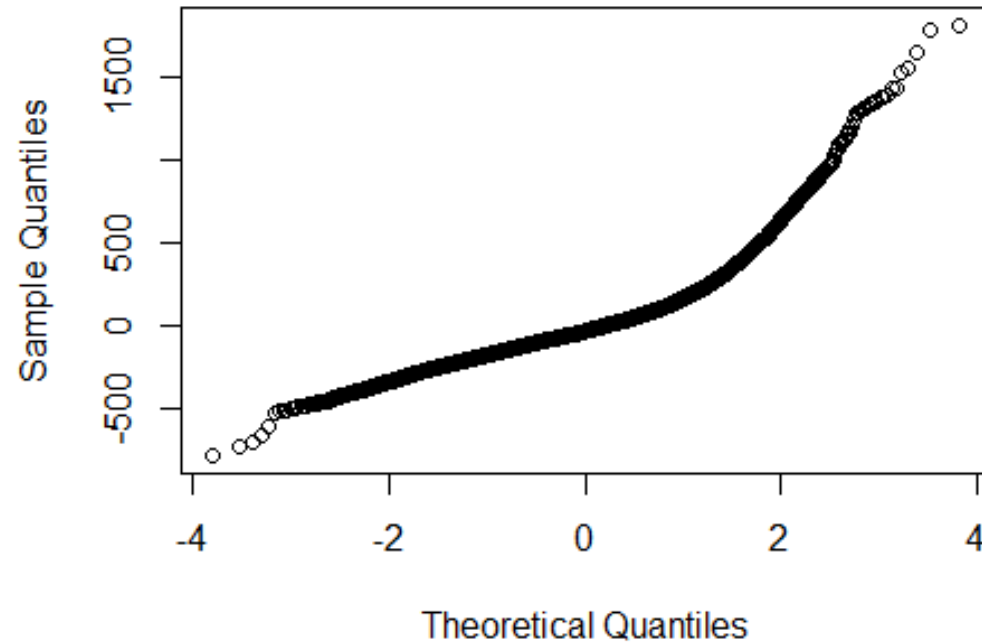
```
PN_Data2 <-  
  PN_Data %>%  
  mutate(trans_RT = 1/sqrt(RT))
```

```
hist(PN_Data2$trans_RT)
```

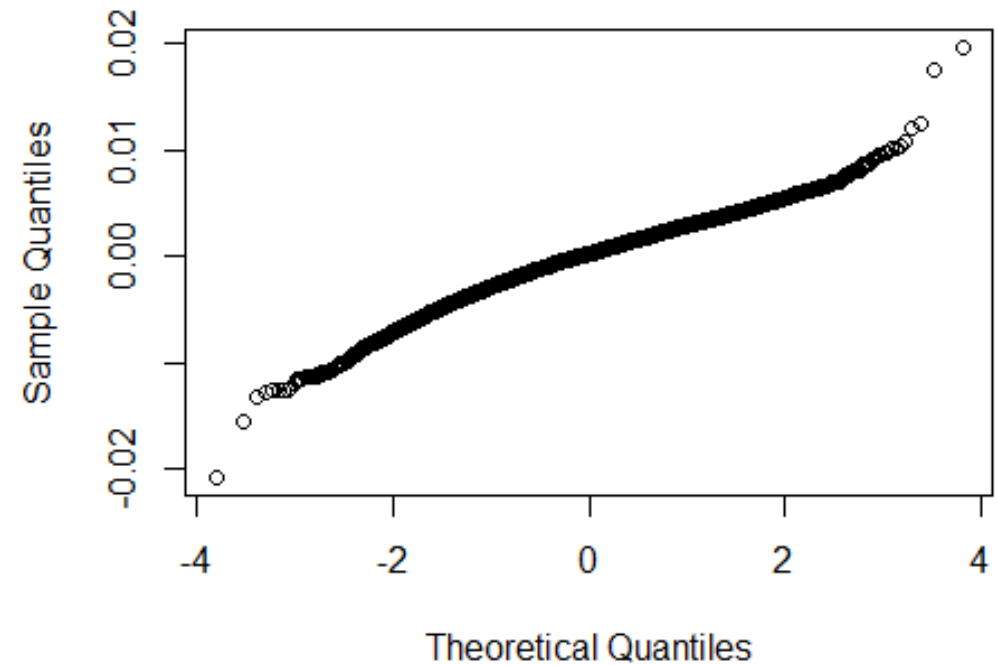


Re-Test: normal distribution of residuals

Normal Q-Q Plot



Normal Q-Q Plot



Multicollinearity

Predictors can be highly correlated and cause problems with the model

Tested with the Variance Inflation Factor (VIF)

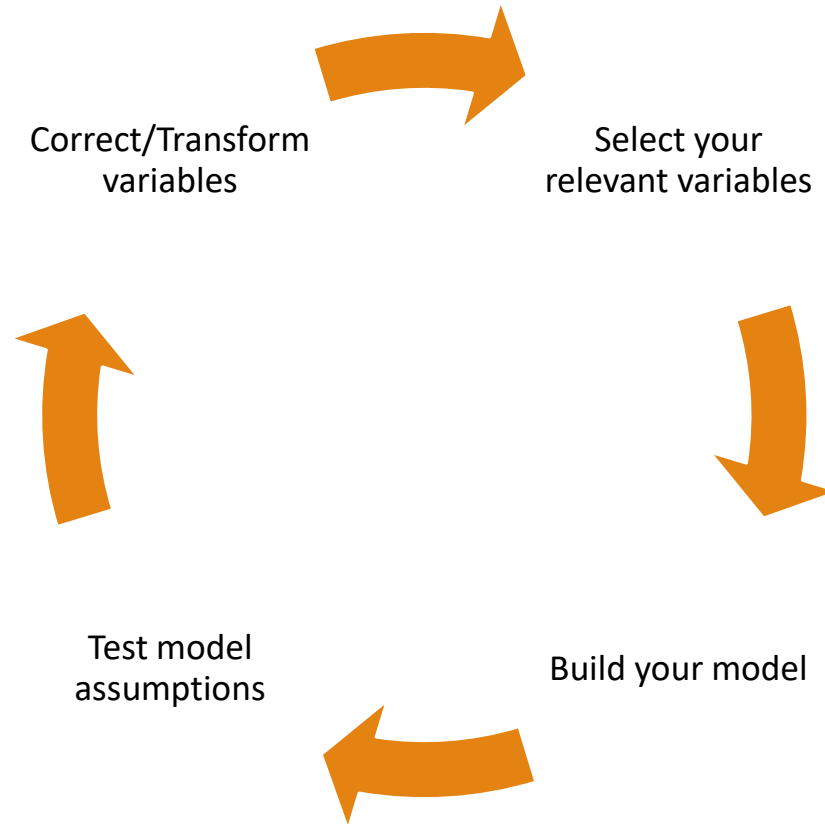
```
library(car)
vif(model_GeneralModel)

##           Group           Context           Age           AoA           Trial
##    1.524975    1.981377    1.274221    1.199493    1.000071
## Group:Context
##    2.218775
```

VIF of 1 indicates no correlation between variables

VIF of 5 or higher might be problematic and coefficient estimates of the model might be highly unreliable

Creating your own models



Example experiment

Bilingual speaker name pictures in a Polish and an English Context

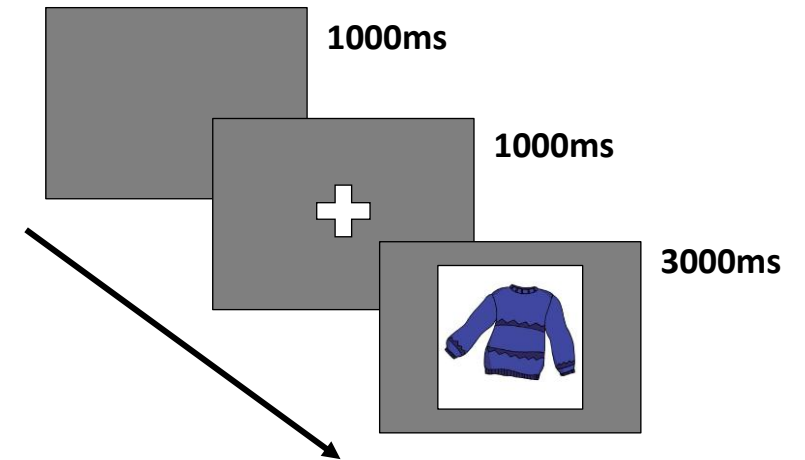
Every speaker saw every picture in both contexts

Experimental variables:

- Language context - UK vs PL
- Group – Experimental vs Control

Grouping factors: Subject and Item

Dependent variable: naming latency



Check the available data

```
head(PN_Data)
```

```
## # A tibble: 6 × 9
```

##	Subject	RT	ItemNr	Group	Context	Trial	lg.freq	Age	AoA
##	<chr>	<dbl>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	AS3008	1049	127	Experimental	UK	4	4.38	36	7
## 2	AW1912	1007	127	Experimental	UK	4	4.38	31	15
## 3	JM2904	794	127	Experimental	UK	4	4.38	39	12
## 4	LM1102	826	127	Experimental	UK	4	4.38	43	15
## 5	MB0509	842	127	Experimental	UK	4	4.38	28	16
## 6	MB2601	1131	127	Experimental	UK	4	4.38	54	15

Check the available data

Dependent variable: RT

Grouping variables:

- Subject and Item

$$RT \sim \text{Context} + (1 \mid \text{Subject})$$

Experimental manipulations:

- Group
- Context

$$RT \sim \text{Group} * \text{Context} + (1 \mid \text{Subject})$$

$$RT \sim \text{Group} * \text{Context} + (1 \mid \text{Subject}) + (1 \mid \text{Item})$$

Background information:

- Age
- Age of L2 acquisition
- Trial
- Word frequency

$$RT \sim \text{Group} * \text{Context} + \text{Age} + \text{AoA} + \text{Trial} + \text{Freq} + (1 \mid \text{Subject}) + (1 \mid \text{Item})$$

$$RT \sim \text{Group} * \text{Context} + \text{Age} + \text{AoA} + \text{Trial} + \text{Freq} + (1 + \text{Context} + \text{Trial} + \text{Freq} \mid \text{Subject}) + (1 + \text{Group} + \text{Age} + \text{AoA} \mid \text{Item})$$



Thank you
for your
attention!
